

Sampling V6 concise report

Choosing a Sampling Method Under Session Caps

Population context: 2,800 sessions across 105 agents, spanning dense_2500, historical_300 corpora with 30 paired seeds. ARM2 uses Azure Search for donor retrieval, and use-case coverage remains provisional because the Maven labels are classification-derived and some assignments are Ambiguous.

POPULATION

2,800 sessions

AGENTS

105 agents

CORPORA

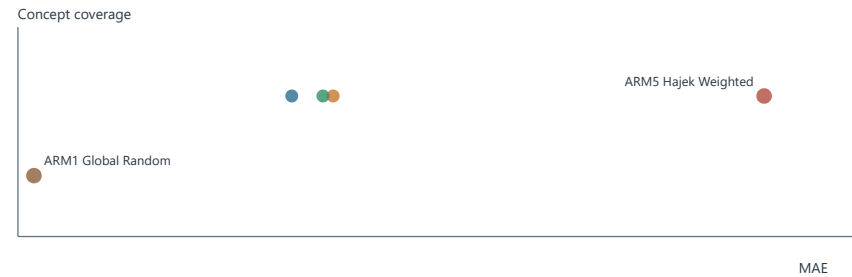
dense_2500 / historical_300

How to read: MAE is the absolute error between the estimator and the full-population benchmark. Concept coverage is the proportion of distinct source concepts in the full population represented by at least one sampled session; higher is better for representation, but it is not the same as MAE or Maven business-use-case coverage.

Decision at a glance

Low-cap / high-cap pattern

At the low-cap end (64 sessions), ARM1 Global Random has the lowest MAE (0.0417) while ARM2 Embedding IDW, ARM3 Agent Round Robin Floor, ARM4 Agent Round Robin, ARM5 Hajek Weighted are tied for broadest concept coverage (13.33%). At the high-cap end (1024 sessions), ARM5 Hajek Weighted remains the best combined MAE/coverage balance at 1024 sessions, so the decision stays conditional: choose the lowest MAE when the budget is tight, but treat concept coverage as the representativeness guardrail when broader source coverage matters more than a small MAE gain.



Selected cap summary

METHOD	CAP	MAE	CONCEPT
ARM1 Global Random	64	0.0417	11.8%
ARM2 Embedding IDW	64	0.1088	13.3%
ARM3 Agent Round Robin Floor	64	0.1169	13.3%
ARM4 Agent Round Robin	64	0.1196	13.3%
ARM5 Hajek Weighted	64	0.2318	13.3%

What is measured

Use the fixed-population evidence correctly

MAE

Lower is better

Absolute difference from the census benchmark at a cap.

CONCEPT COVERAGE

Higher is broader

Share of distinct source concepts in the full population that are represented by at least one sampled session.

LIMIT

Not general inference

This is descriptive evidence for this synthetic population and cap grid, not a claim about all sessions everywhere.

Five techniques

Selection and estimator are not the same

ARM1

Steps	Draw sessions uniformly across the fixed population; keep the selected set simple and budgeted.
Estimator	Unweighted sample mean for the pass-rate estimate.
Strength	Simple, robust low-cap baseline with minimal tuning.
Risk	Can miss concept-rich regions when a few hot agents dominate the population.
Choose when	Use as the clean baseline when the goal is speed and low complexity under a strict cap.

ARM2

Steps	Use embedding similarity and the Azure Search donor pool, then apply same-agent IDW for unjudged units.
Estimator	Same-agent IDW plus donor-weighted adjustment rather than a raw mean.
Strength	Local fidelity and calibration when similar sessions are available.
Risk	Embedding or search quality can dominate the result and amplify duplication risk.
Choose when	Choose this when donor quality matters and the objective emphasizes explicit imputation quality.

ARM3

Steps	Enforce a floor to protect agent coverage, then allocate the remaining budget across Maven strata.
Estimator	Unweighted mean across the selected sessions.
Strength	Broad representation and a minimum floor before the cap is consumed.
Risk	Some of the cap is spent on coverage protection instead of pure MAE minimization.

ARM4

Steps	Round-robin across agents and then balance by Maven strata without a hard floor.
Estimator	Unweighted mean.
Strength	Good agent diversity and concept spread under the cap.
Risk	Use-case strata can still miss rare concept pockets if the strata are noisy.
Choose when	Choose it when diversity and representation matter more than an exact floor guarantee.

Choose when

Choose it when broad representation is a non-negotiable requirement.

ARM5

- Steps**

Use the same membership as ARM4, but switch the estimator to a Hajek ratio estimator.
- Estimator**

Weighted ratio estimate that respects selection propensities.
- Strength**

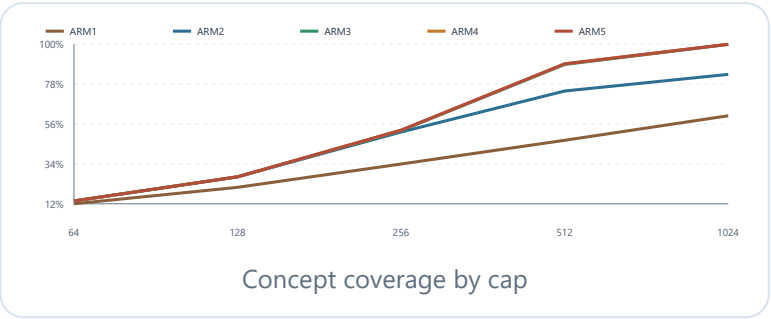
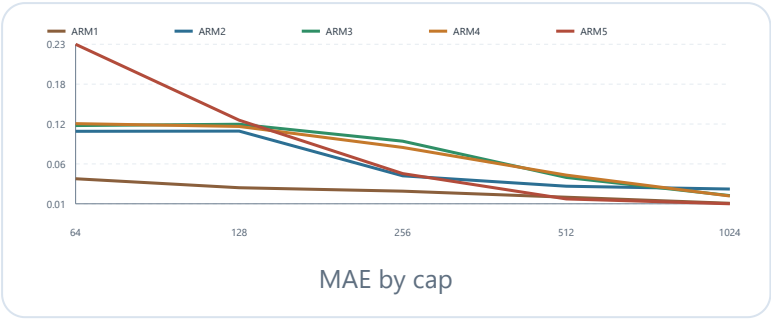
Preserves ARM4 coverage while improving estimator design alignment.
- Risk**

Low-cap weighting can be unstable and slightly raise MAE.
- Choose when**

Choose it when you want ARM4-style coverage with a more design-aware estimator.

Results story

Visual-first view



	64	128	256	512	1024
Lowest MAE	ARM1 0.0417	ARM1 0.0290	ARM1 0.0241	ARM5 0.0132	ARM5 0.0063
Most concepts	ARM2-ARM5 13.3%	ARM2 / ARM4 / ARM5 26.7%	ARM4 / ARM5 52.4%	ARM4 / ARM5 88.9%	ARM4 / ARM5 99.8%

Each column names the leader; ties are preserved.

Cap-by-cap leaders: lower MAE and higher concept coverage

Takeaway: the best method changes by cap and objective. Low-cap comparisons are most sensitive to the estimator choice, while high-cap patterns let broader representativeness and lower MAE coexist without claiming a universal winner.

Conclusion

Conditioned on the fixed bundle

Recommendation: If the objective is a tight-cap, low-error baseline, prefer the method with the lowest MAE at the active cap. If the objective is representativeness across source concepts and agents, favor the method with the highest concept coverage at that cap, while acknowledging that concept coverage is not accuracy and not a proxy for Maven business-use-case coverage. On this bundle, the pattern is clear: **low-cap** decisions are driven mainly by MAE, while **high-cap** decisions can lower MAE and increase concept coverage without implying a single universal winner.